Session 1 — Getting Started

1. The User Interface

Before we jump into making sounds, let’s take a quick look at the user interface.

When you first load up a new Ableton project, you’ll see something like this:

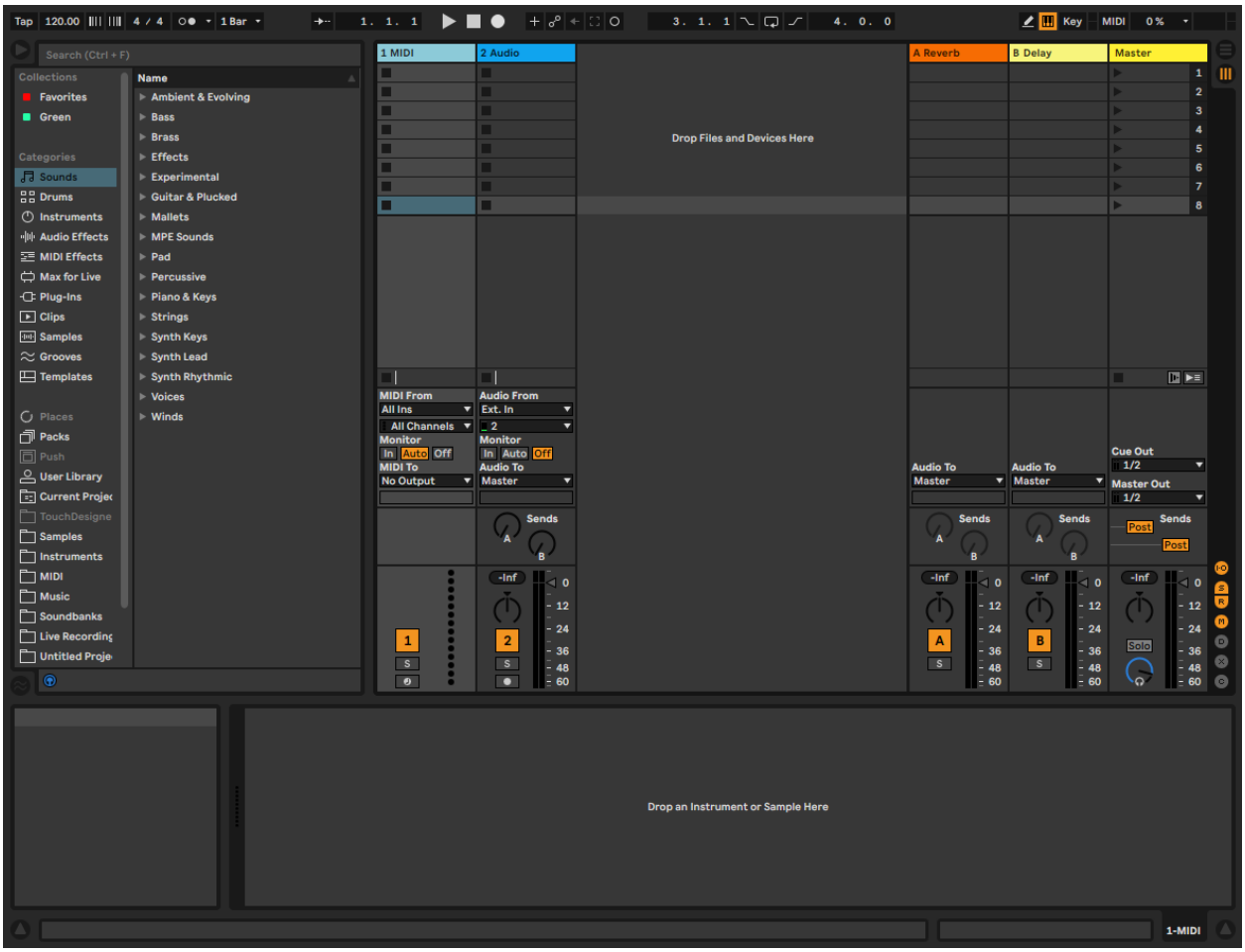


fig. 1 A fresh Ableton Project

The UI can be separated into the following regions:

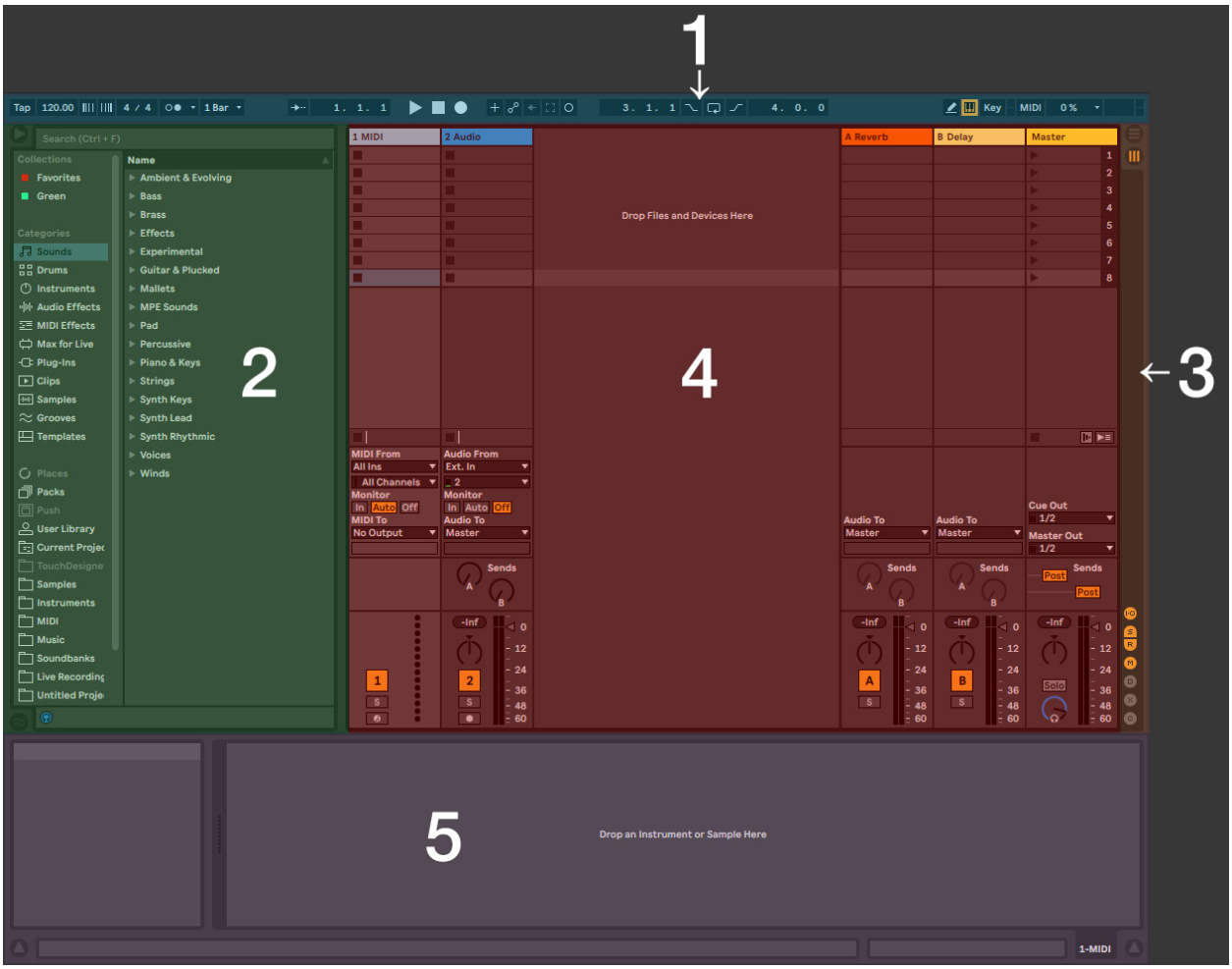


fig. 2 The regions of Ableton's UI

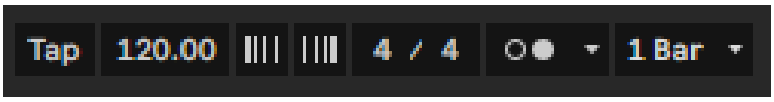
In brief, these are:

1. **Toolbar** — Tempo, playback control, and CPU stats
2. **Browser** — Access files and devices
3. **Sideba**r — Switch views in the clip area and toggle track information
4. **Clip Area** — Where most of the work gets done. Can be in either session or arrangement view
5. **Instrument/Clip View** — Modify instruments, samples, or MIDI and get information

1. Toolbar

The toolbar can be broken into four main sections. From left to right these are:

Tempo & Quantization

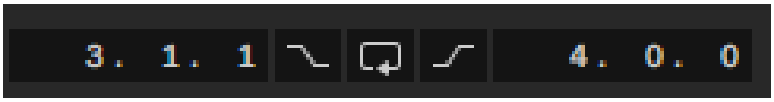


These settings allow you to set and control the global tempo of your song, sync tempo to external audio, change the time signature of your song, enable the metronome, and change quantization length.

Recording, Playback, & Automation

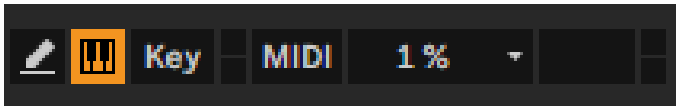
These settings let you control audio playback, recording, and automation settings.

Looping



These settings let you monitor and change looping settings.

MIDI & CPU



Here you can activate draw mode, toggle MIDI via computer keyboard (shortcut: **M**), and map your MIDI controller to settings within Ableton, and monitor CPU load.

2. Browser

The browser is where you'll access all of your devices, instruments, FX, files, and other presets within Ableton. This is separated into **Collections** where you can organize and color code individual files, **Categories** which features all built-in files and devices, and **Places** which features user files.

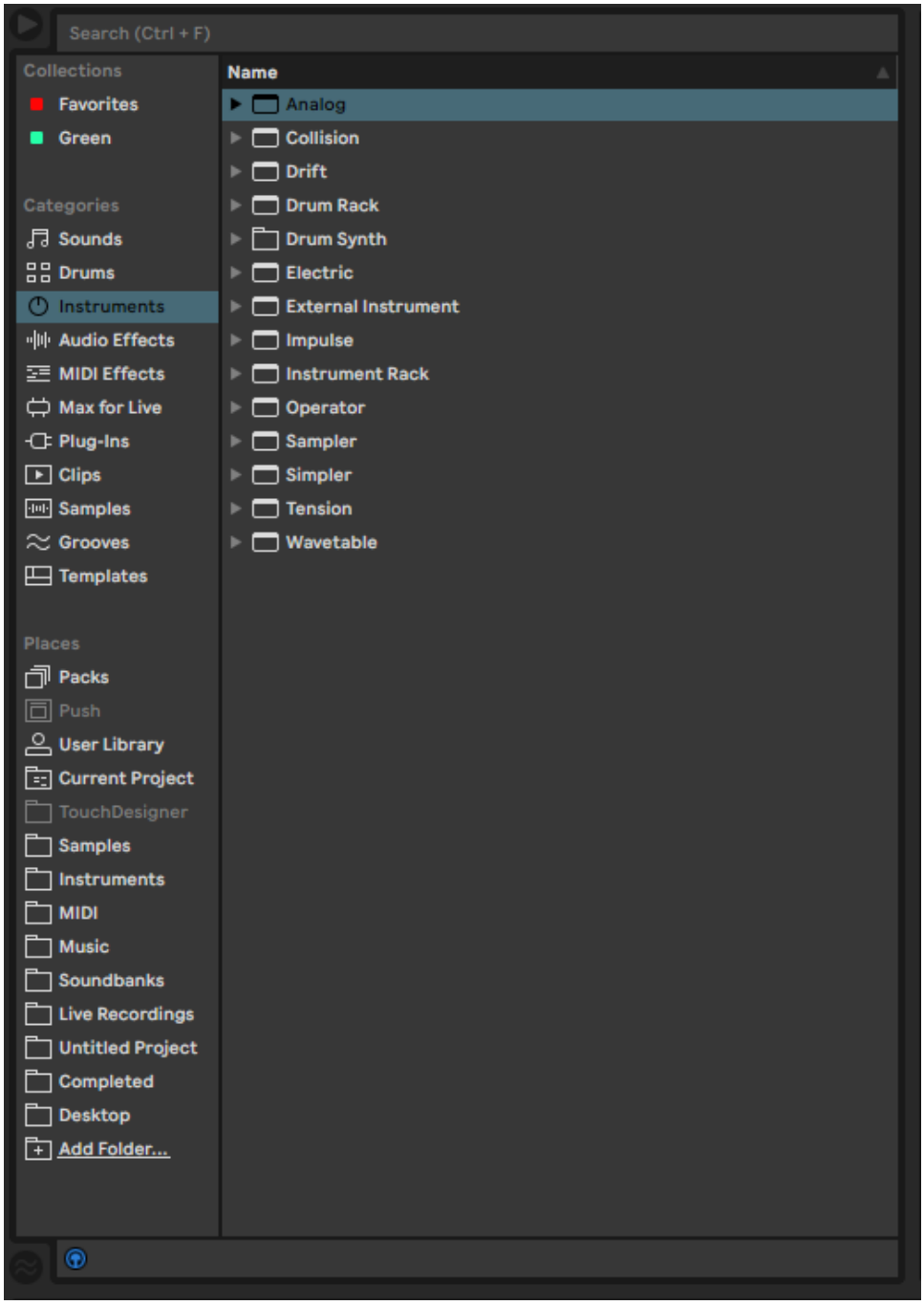


fig. 3 The browser window

By clicking **Add folder…** at the bottom of Places, we can add new folders that we can access within our project. I would recommend adding the **Live Recordings** folder, which contains all Ableton projects you make. This allows you to then easily drag and drop anything from one project file to another. The default file path for Live Recordings on Windows is:

```
C:\Users\[Username]\Documents\Ableton\Live Recordings
```

For Mac, the default file path is:

```
HD/Users/[username]/Music/Ableton/Live Recordings
```

If you have a folder of audio samples, I would also recommend adding this to Places as we did above.

3. Sidebar

The sidebar has **two main sections.**

1. **Arrangement/session view selector**

   We'll look more at this in the next UI section.

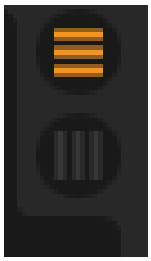


fig. 4 The selector. Arrangement view is on the top, session view is on the bottom

2. **Track information**

   These buttons show/hide information on each track of audio or MIDI. From top to bottom these are: the In/Out section, returns, the mixer, and delays.

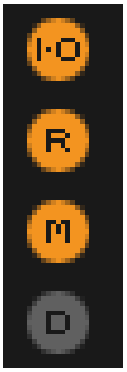


fig. 5 Toggle each button to show/hide their respective section on each track

4. Clip Area

The clip area is the **primary area** you'll be working in while using Ableton. The clip area has two different views that can be switched between by pressing TAB or clicking the selectors in the sidebar.

Session View

fig. 6 Session view with a MIDI and audio track, as well as reverb and delay buses

Session View consists of audio or MIDI **tracks** arranged in **columns,** and audio or MIDI **clips** arranged in **rows**. You can think of tracks as individual instruments, and clips as bits of sound or notes being played on each instrument. We'll dive more into this concept soon.

Session view is great for:

- DJing
- Getting down starting ideas
- Gathering a collection of different arrangements or samples
- Auditioning different ideas and arrangements together
- Generative music with Follow Actions (we'll be covering this in the next session)

Arrangement View

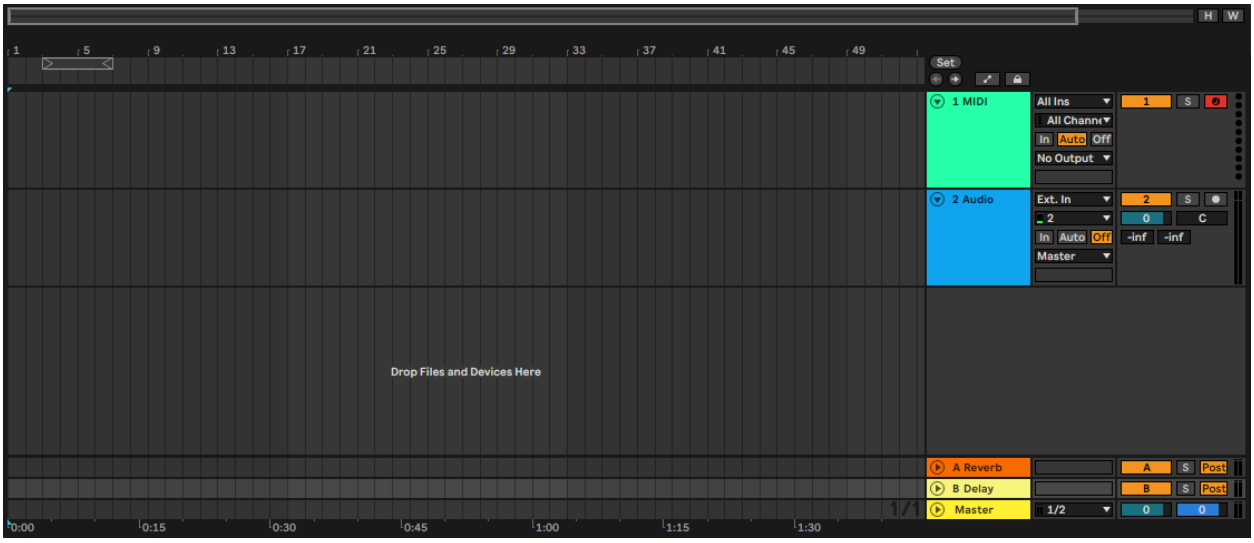


fig. 7 Arrangement view.

Arrangement view consists of the same audio or MIDI tracks as we just saw in Session view, but now laid out in a single **timeline**. This might be familiar to you if you've worked with other DAWs, or other software such as Adobe Premiere.

Arrangement view is great for:

- Composing multiple instruments or audio tracks together
- Automating parameters
- Arranging a finished track

5. Instrument/Clip View

The other area in Ableton we'll be spending most of our time in is the instrument/clip view. This consists of two sub-windows.

On the left is **info view**. This window will be your best friend while learning Ableton, as it will display information about every window and parameter you hover your mouse over. This includes every element of the UI as well as features and parameters in Ableton's built-in instruments and effects. This window can be hidden by clicking the arrow in the bottom left.

To the right is the **instrument/clip view**. The instrument view is default, and will show you the effects chain of any track you have highlighted. When an audio clip is double-clicked, this window will change to the clip view where you can see detailed controls of your audio file. These views can be switched with ⇧ **+ TAB.**

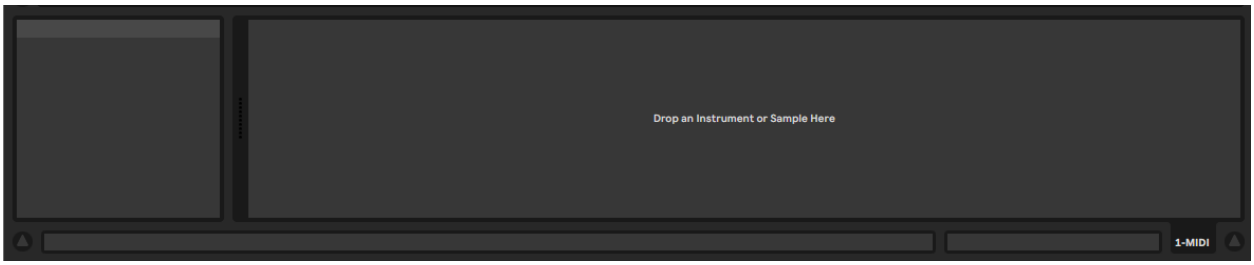


fig. 8 Info view on the left, and instrument view on the right

2. Getting Sound

Now that we know our way around Ableton, let's make some noise. First we're going to look at working with audio, then after we'll take a look at MIDI.

i. Audio

a. Audio Clips

Audio clips can be **imported** into Ableton by dragging and dropping from either a folder on your computer or a Places folder in the Ableton Browser into the clip area where it says "Drop Files and Devices Here". Notice this creates a new audio track, with the file we just dragged in as the first track.

To **record** audio into Ableton, we first need to access the in/out controls on the audio track we want to record into (remember, these controls can be shown/hidden via the sidebar). In a new audio clip, click the dropdown menu underneath "Audio From" and select either an external device → Ext. In or audio from another audio or MIDI track in our project → name of track.

To hear the audio coming in, click the In button under "Monitor" on the track you're recording into. Finally, make sure the track has recording armed by clicking the arm button at the bottom of our audio track. Then just click the record button in the clip you want to record into and click stop when you're done.

fig. 9 How to configure an audio track to record audio

Try double clicking the audio clip. This will open our audio in the **clip view** and allow us to work directly with the parameters of our audio file. This view is broken up into two sub-windows. The sub-window on the left displays all the controls for our audio clip.

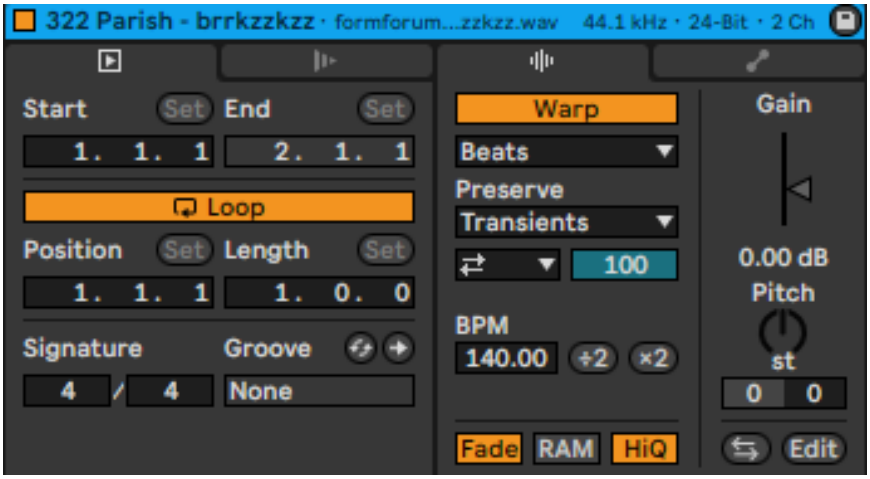


fig. 10 The controls of an audio clip

On the left we have controls for looping our audio clip when it finishes. This can be turned on/off with the loop button, and the length can be set with the appropriate Start, End, Position, and Length parameters.

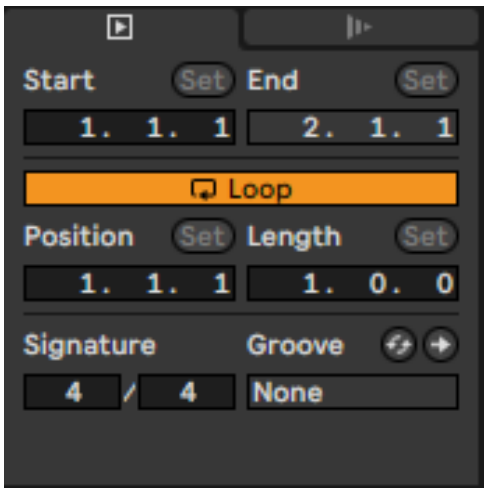


fig. 11 Looping controls of an audio clip

On the right, we have controls for warping (we'll get to this in just a moment), the estimated BPM of the clip, as well as controls for controlling gain, pitch, and reversing on the right.

fig. 12 Audio clip parameters, including warp control (left) and gain, pitch, and reverse (right)

To the right is the waveform of our audio clip. As we adjust parameters on the left, we can see the changes in the waveform on the right.

fig. 13 Audio waveform display

With the basics out of the way, let’s move on to learning to harness one of Ableton’s most unique and powerful features for sound design — warping.

b. Warping

Warping is Ableton's method of allowing audio clips of any BPM to stay in tempo with the global BPM of the track, even if those audio clips are manipulated in some way. This works by stretching or compressing the transients (think: the peaks you see in the waveform) of the audio clip. A **warp marker** is like a tag that tells Ableton where a transient is.

When Warp Mode is turned on, Ableton will automatically analyze your audio file for and assign warp markers where it finds **transients**. These points are indicated by the small gray downward arrows directly above your waveform when warping is on. You can double click in this region to place a marker yourself, indicated by the yellow box above the arrow.

In ordinary usage, you can place these warp markers wherever you see a transient to ensure your audio clip sounds as true to the original in rhythm and texture regardless of your project's global BPM.

For our purposes, we can use the variety of warp modes to create exciting new sounds from old ones.

fig. 14 A drum loop with automatic warp markers (gray arrows) and manual warp markers (yellow arrows)

Getting Weird with Texture Mode

Texture mode is a warping mode that utilizes ***granular*** resampling.

The parameter “Grain Size” determines how small each grain the sample is broken up into are. Smaller values produce glitchier more textural sounds. The “Flux” parameter applies randomization to the length of these grains, with larger values producing a jittery or watery quality.

In normal usage, texture mode is best for harmonically complex sounds like chords, while the similar Tones mode is best on monophonic material.

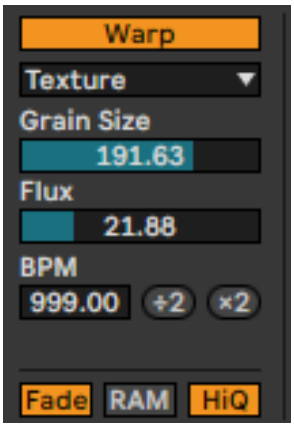


fig. 15 Grain Size, Flux, and BPM in Texture Mode

Texture mode produces interesting results when we abuse Ableton’s warp markers. By placing markers **off-beat**, or on non-rhythmic audio clips, then spread them around our clip, we can produce exciting new textures from our original audio clip.

Try playing with the **BPM**, **Grain Size**, **Flux**, and **Pitch** parameters in your audio clip to create new rhythms, textures, and timbres. The larger the difference between the global BPM and clip BPM, the more drastic the results (i.e: 20 global BPM and 999 clip BPM produces the greatest amount of warping).

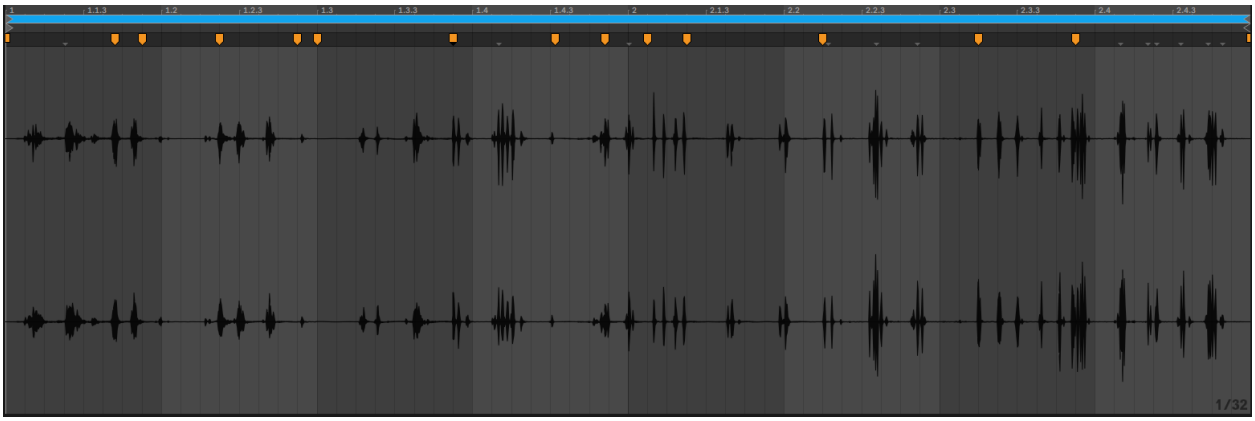


fig. 16 Warp markers placed arbitrarily on an audio clip

Complex Pro is another warp mode that can produce similarly interesting results as Texture Mode but typically with a smoother (less textural) quality and a more metallic sound. Try experimenting with using one warp mode, resampling the audio to a new audio clip, then using another mode to warp it again.

c. Automation

Unlocking Warp Mode's Power with Clip Automation

While manually tweaking the parameters in texture mode is fun, we can unlock much more precision and power in warp mode by automating these parameters instead. To do this, click the last tab in our clip to open the **Envelopes** section.

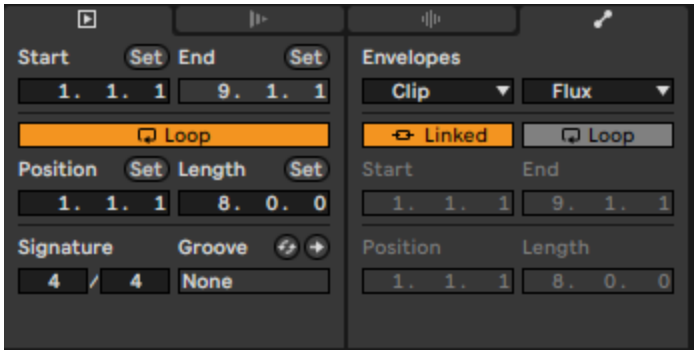


fig. 17 The clip menus. Envelopes can be accessed from the last tab.

Here we can select parameters pertaining to our audio clip from the first two dropdown menus, with the right menu selecting the parameter we're automating. A blue dotted line

will appear on our waveform. This is our **automation curve**, which is measured in percent in clips (as opposed to the value of the parameter in arrangement automation as we'll see later). This is 100% by default.

For example, if we select Grain Size, we are at 100% of the value for Grain Size that we set earlier, and we can automate this value by drawing in lower values in our automation curve. This can be done by clicking on the line of the curve to place a node. Click and drag this node to change its value. Mouse over a portion of the line and click and drag while holding ALT to draw **inflection curves** in our line.

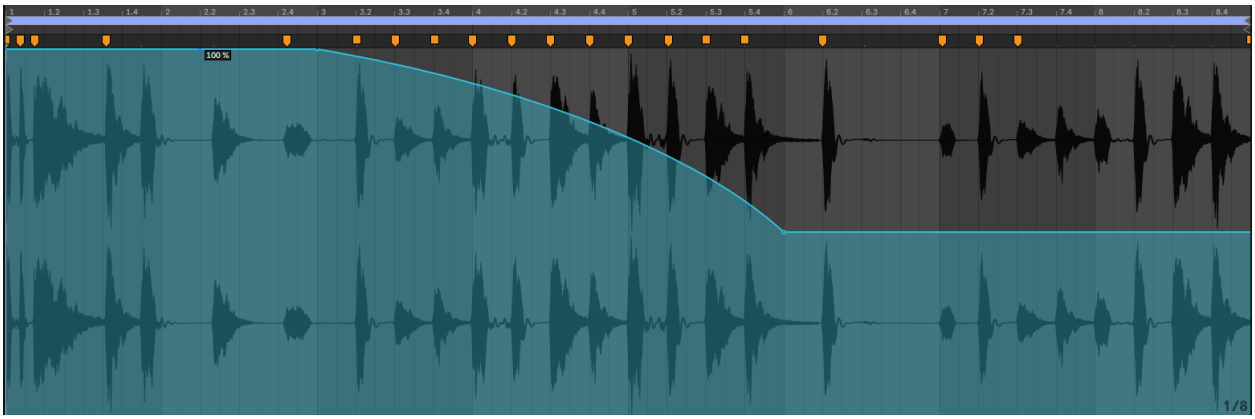


fig. 18 An automation curve made by placing nodes and using ALT to alter the inflection of the line.

Another way we can do this is in **draw mode**. Press B or right click on the waveform and select "Draw Mode" to access it. The mouse cursor will be replaced by a pencil. Click and hold to draw the automation curve with your mouse. This is best done with the grid turned off for maximum precision: right click the waveform and under "Fixed Grid:" select "Off".

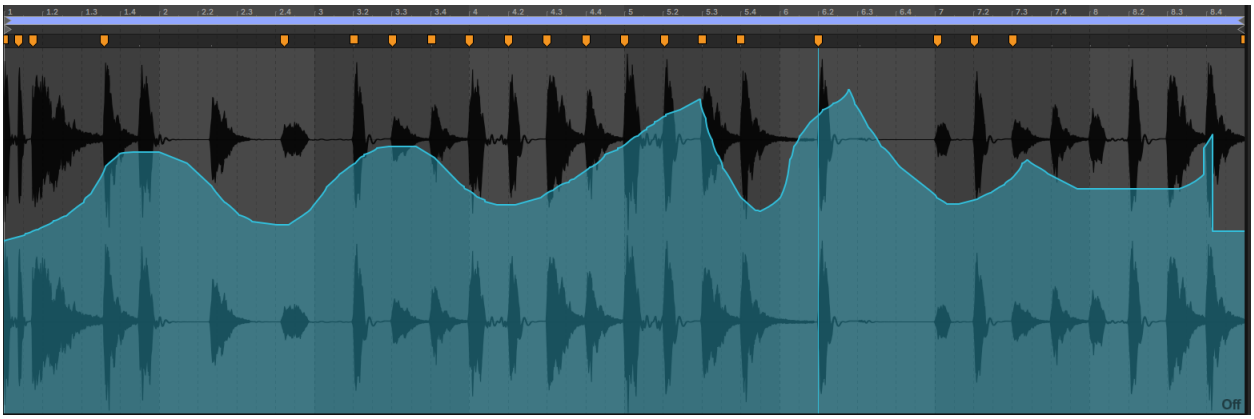


fig. 19 An automation curve made in draw mode with the grid turned off.

Each parameter has its own automation curve. Try selecting multiple parameters from earlier, such as Grain Size, Flux, and Transposition (pitch) and draw unique curves for each to unlock wildly new dynamic sounds from a simple audio clip.

ii. MIDI

b. MIDI Clips

Just like audio clips, MIDI clips can be imported by dragging and dropping. However, we can also make a new MIDI clip by double clicking in an empty clip in a MIDI track. This will pop up the MIDI Clip view for a new 1 bar MIDI clip. Like the clip view for audio, there are two sub-windows. The first is the clip editor.

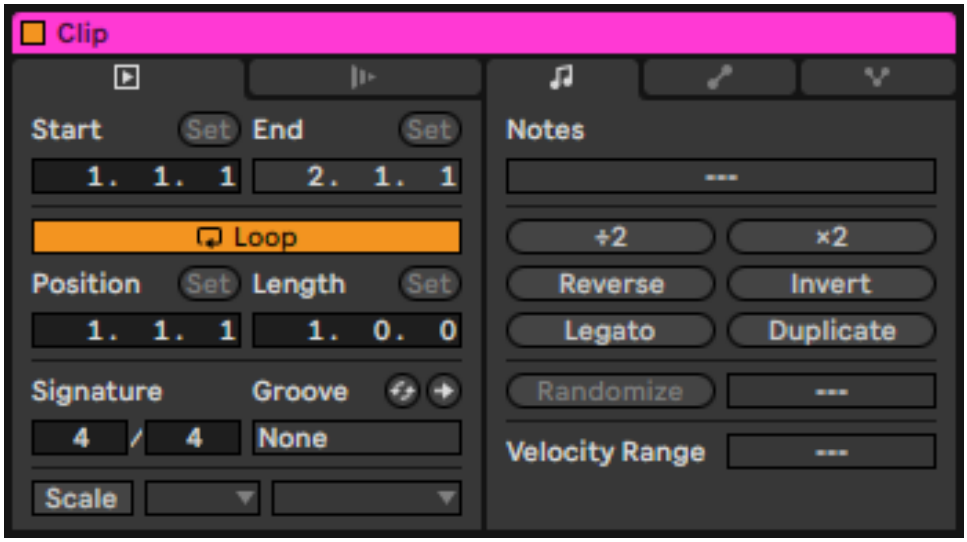


fig. 20 MIDI Clip Editor

This contains the same looping and envelope controls as an audio clip. What's different here is the **Notes tab** on the right. Here we can use %2 or x2 to shorten or lengthen notes respectively, as well as reverse, invert, and add legato. We can also see a Randomize button and Velocity Range, which will randomize the velocity of any played notes accord to the range we set.

The other sub-window is the MIDI Note Editor, where we can draw or play in notes into our clip.

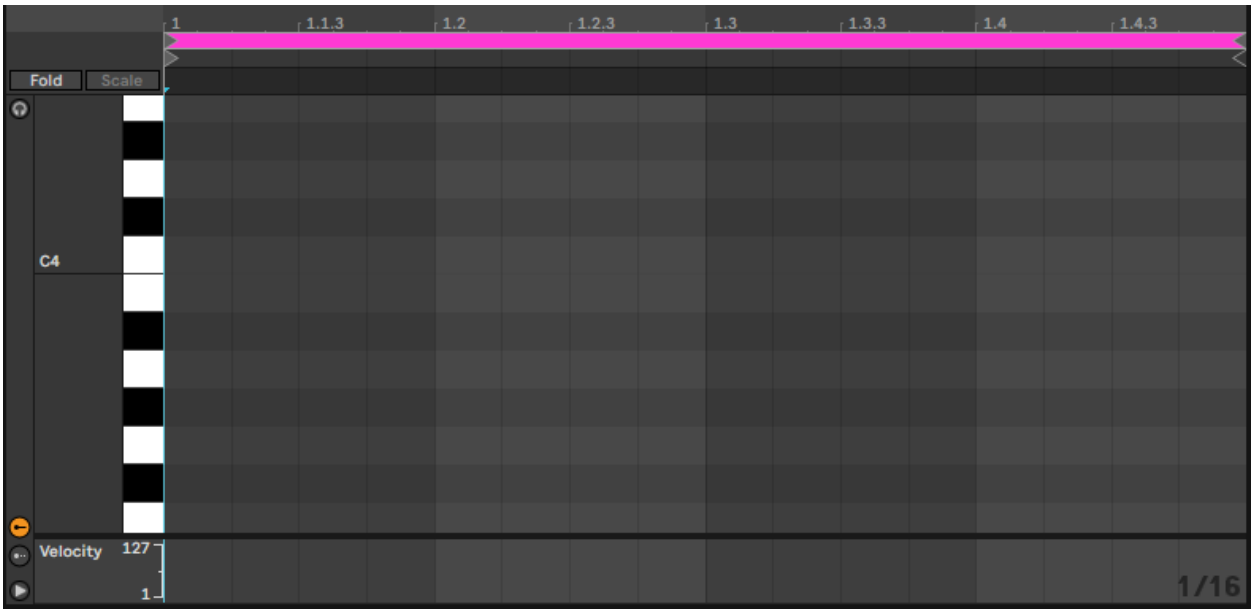


fig. 21 MIDI Note Editor

This editor acts as a grid with the x-axis corresponding to time and note values being displayed by the piano roll along the y-axis.

Notes can be added by double clicking, or by recording in notes from your keyboard or a MIDI controller. To allow your computer keyboard to be used to play notes, press **M** to enable the Computer MIDI keyboard. Then simply click record on an empty MIDI clip as we did with audio and press any any keys along the ASD row on your keyboard for white keys and corresponding notes on the QWE row for black keys.

You can modify notes using your mouse to click and drag its length in either side or drag to different notes on the piano roll.

Directly underneath the piano roll is the **Velocity Editor**, which displays the velocity, or how hard the note is being played, for every note in the piano roll above it. You can click and drag these values to whatever velocity you want (I recommend expanding the velocity editor by dragging its top edge up).

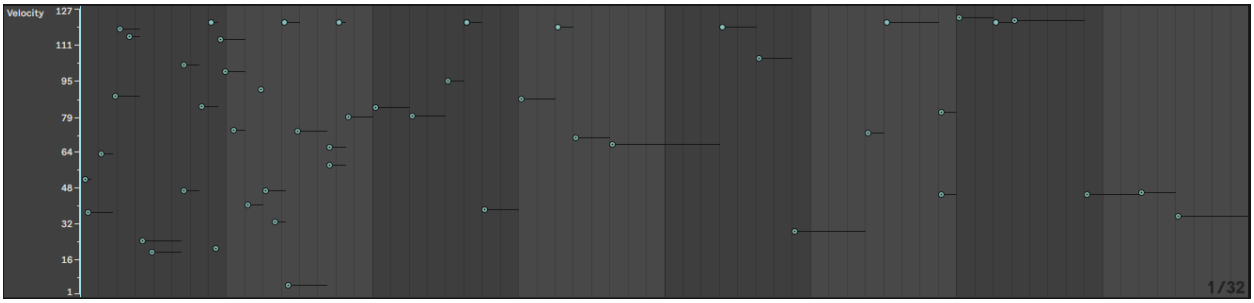


fig. 22 Velocity Editor

If you want to add a human touch to your compositions, highlight all your notes with CTRL + A, then drag the Random Range next to the Randomize button up. A good range for a subtle touch is 15-40 depending on the instrument. Then simply click **Randomize** to give variation to the velocity of all your notes.

Using Fold to Create Scales

The **Fold** button can be used to hide any notes on the piano roll that don't have a corresponding note in the editor. We can use this as a neat trick to display only notes in a scale we want to use in the MIDI Note Editor. To do this, click in the notes of your desired scale. Here I'll use B Minor.

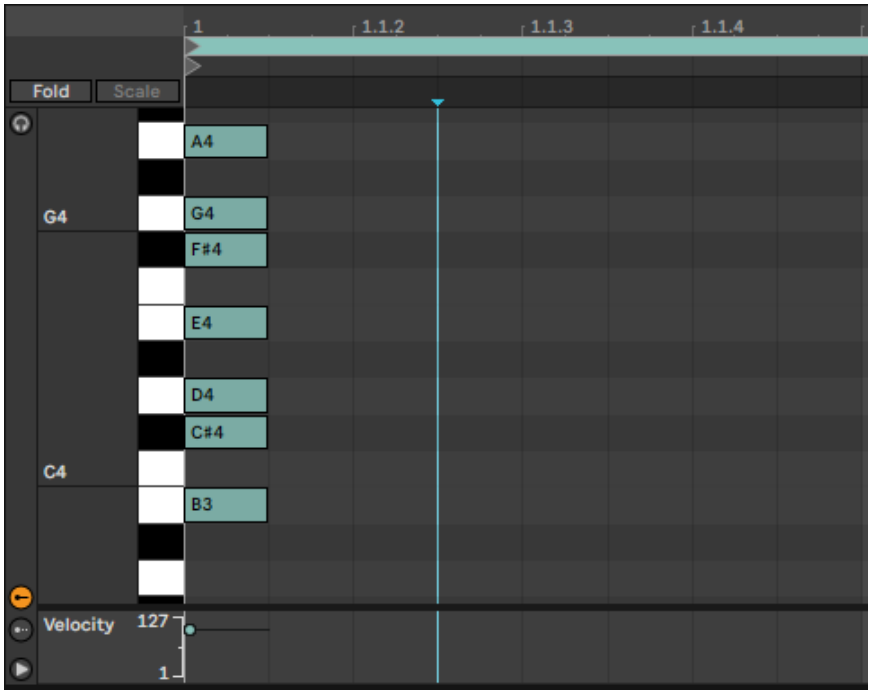


fig. 23 B Minor Scale in one octave

To expand the scale to multiple octaves, press CTRL + A to highlight all of our notes, CTRL + D to duplicate them, SHIFT + ↑ to move all notes up an octave, then ← to move them back to the start. Do the same thing again but use SHIFT + ↓ to move to lower octaves.

Finally, press CTRL + A again to grab all of our notes, and click and drag them to the left until our notes go behind our start marker. Now click the Fold button. Because these notes are outside our start marker, they won't be played, but instead will take advantage of Fold to only show notes in scale. You can now click and play around to your heart's content knowing that it will all be in scale!

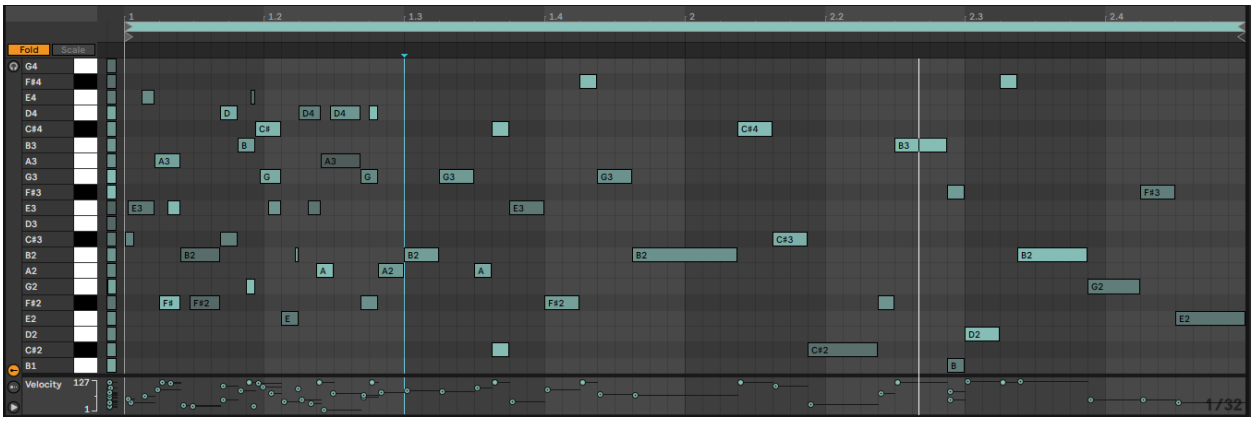


fig. 24 A fully random composition in B Minor made using the Fold trick